

Department of VLSI Design and Technology

Curriculum Summary - R2024

A semiconductor-focused programme pathway with foundations, design practice, verification, implementation, and professional readiness.

Semester I: Engineering mathematics, applied physics, programming fundamentals, basic electrical and electronics concepts, communication skills.

Semester II: Engineering mathematics, network theory, electronic devices, digital systems foundations, problem-solving practice.

Semester III: Signals and systems, analog circuits, digital electronics, data structures, semiconductor fundamentals.

Semester IV: Microprocessors and microcontrollers, control systems, communication fundamentals, CMOS concepts, laboratory practice.

Semester V: VLSI design fundamentals, HDL-based design, embedded systems, digital signal processing, design laboratory.

Semester VI: Physical design concepts, verification methods, FPGA-based design, static timing analysis, project-based learning.

Semester VII: Advanced VLSI electives, industry-oriented laboratory work, research methodology, major project phase I.

Semester VIII: Major project phase II, internship or industry exposure, seminar, and professional readiness.

Elective Basket: Digital Signal Processing, Artificial Intelligence for VLSI, Internet of Things, Machine Learning, Advanced CMOS Technology, FPGA-Based System Design, System-on-Chip Design, MEMS Technology, Nano Electronics, Embedded Linux, RTL Design and Verification, Static Timing Analysis, Physical Design Automation, Semiconductor Packaging Technology